

Jia Wei TEH

CURRICULUM VITAE

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Research Interests

Stellar feedback and the evolution of interstellar bubble in giant molecular clouds. Lead developer of TRINITY, a 1D thin-shell feedback code, and its application with resolved galaxy surveys (PHANGS, SDSS-V LVM, JWST).

Education and Research Experiences

PhD in Astronomy and Cosmology Jun 2022 - present
International Max Planck Research School for Astronomy and Cosmic Physics
University of Heidelberg, Heidelberg, Germany
Supervised by: Prof. Ralf Klessen, Prof. Simon Glover, Dr. Kathryn Kreckel
Expected defence: spring 2027

BSc. (Adv.) (Hons.) in Astronomy & Astrophysics Feb 2018 - Nov 2021
(Equivalent to UK four-year integrated MSci degree)
Australian National University, Canberra, Australia

- Graduated with First Class Honours
- Thesis: *Constraining the escape fraction with LEGUS–SIGNALS survey*
- Supervised by Dr. Kathryn Grasha, Prof. Mark Krumholz, Dr. Andrew Battisti

Estimating masses of galaxy clusters with deep learning Nov 2021 - Feb 2022
CSIRO Space and Astronomy Summer Research Studentship
Supervised by Dr. Nikhel Gupta

Beyond the face value of HCN emission in Antennae Jun 2021 - Aug 2021
The Leiden/ESA Astrophysics Program (LEAPS), Leiden Observatory
Supervised by Dr. Ashley Bemis

Can black hole binaries explain strong emission lines in galaxies? Nov 2020 - Feb 2021
Centre for Astrophysics & Supercomputing Summer Research Studentship, Swinburne U.
Supervised by Dr. Themiya Nanayakkara

Resolved study of star clusters and H II regions in NGC 628 Jan 2020 - Jul 2020
Undergraduate Research Experience, Australian National University
Supervised by Dr. Kathryn Grasha, Dr. Andrew Battisti

Refereed Publications

Teh, J.W., Klessen, R.S., Glover, S.C.O., Kreckel, K., 2026. *TRINITY: A coupled model of winds, radiation, and photoionised gas in molecular clouds. I. Methods and validation.* Astronomy & Astrophysics. doi:10.1051/0004-6361/202661074

Teh, J.W., Grasha, K., Krumholz, M.R., Battisti, A.J., Calzetti, D., Rousseau-Nepton, L., Rhea, C., Adamo, A., Kennicutt, R.C., Grebel, E.K., Cook, D.O., et al., 2023. *Constraining the LyC escape fraction from LEGUS star clusters with SIGNALS H II region observations: A pilot study of NGC 628.* Monthly Notices of the Royal Astronomical Society, 524(1), pp.1191-1210.

Scheuermann, F., Kreckel, K., **Teh, J.W.**, Belfiore, F., Groves, B., Barnes, A.T., et al., 2026. *Stellar associations powering H II regions: II. Escape fraction of ionising photons.*

Astronomy & Astrophysics, 708, A202.

Polak, B., Mac Low, M.-M., Klessen, R.S., **Teh, J.W.**, Cournoyer-Cloutier, C., Anderson, E.P., Appel, S.M., et al., 2024. *Massive star cluster formation: I. High star formation efficiency while resolving feedback of individual stars*. Astronomy & Astrophysics, 690, A94.

Soler, J.D., Beuther, H., Glover, S.C.O., Klessen, R.S., Ott, J., Rugel, M., **Teh, J.W.**, et al., 2026. *The Neutral Atomic Hydrogen in the solar neighbourhood (NeAtHood) project: I. Ghost in the shell: neutral atomic hydrogen in the extended Orion nebula*. Astronomy & Astrophysics, 711, A85.

Sheriff, K.E., Mills, E.A.C., Mai, X., Siwakoti, U., Duval, S.E., Bolatto, A.D., Levy, R.C., **Teh, J.W.**, et al. *JWST Observations of Starbursts: A Young Bubble in NGC 253's Central Starburst*.

Maconi, E., Soler, J.D., Reissl, S., Girichidis, P., Klessen, R.S., Hennebelle, P., Molinari, S., Testi, L., Smith, R.J., Sormani, M.C., **Teh, J.W.**, 2023. *Modelling Local Bubble analogs: synthetic dust polarization maps*. Monthly Notices of the Royal Astronomical Society, 523(4), pp.5995-6010.

In Preparation **Teh, J.W.** and the LVM collaboration. *Stellar feedback in the Rosette Nebula*.

Teh, J.W., Klessen, R.S., Glover, S.C.O., Kreckel, K. *TRINITY II: a parameter survey of giant molecular cloud evolution under stellar feedback*.

Guevara, C., Kabanovic, S., Schneider, N., Dannhauer, S., García, P., Simon, R., Kreckel, K., Ossenkopf-Okada, V., **Teh, J.W.** *The SOFIA bubble project*

**Collaborations
and Memberships**

- PHANGS collaboration
- SDSS-V / Local Volume Mapper
- ECOGAL ERC Synergy consortium
- European Astronomical Society (EAS)
- Astronomische Gesellschaft (German Astronomical Society)
- Notice of Intent, *Tracing the Origins of Multiphase Outflows in Galactic Star-Forming Regions to Define Habitable Worlds Observatory Spectroscopic Requirements*; NASA Habitable Worlds Observatory Precursor Science Investigations, PI J. L. Pineda (JPL) 2026

Software

TRINITY: 1D thin-shell stellar feedback code for interstellar bubble evolution.

Selected Talks

- Aug 2026: Cologne–Bonn Symposium on the Habitats of Massive Stars across Cosmic Time, University of Cologne, DE
- Jun 2026: *Invited*, “How to Effectively Use Colours”, early-career session, European Astronomical Society Annual Meeting, Lausanne, CH
- 2026: SDSS-V Collaboration Meeting, KU Leuven, BE
- Nov 2025: “Colour Theory and Visualisation”, HGSFP Winter School, Obergurgl, AT
- Jun 2023: 2nd IMPRS Seminar, Max Planck Institute for Astrophysics, DE
- Nov 2022: 1st IMPRS Seminar, Heidelberg University, DE
- Oct 2022: Local Volume Mapper Early Science Workshop, DE

	• Feb 2022: Student Symposium, CSIRO, Marsfield NSW, AU • Dec 2021: Australian Institute of Physics Summer Meeting, Queensland University of Technology, AU • Nov 2021: Mt. Stromlo Student Seminars, Australian National University, AU • Oct 2021: Galaxy Lunch Talk, RSAA, Australian National University, AU • Sep 2021: Australasian Council of Undergraduate Research Conference, Australian National University, AU	
Posters	• Jun 2026: European Astronomical Society Annual Meeting, Lausanne, CH • Jul 2025: 18th Potsdam Thinkshop, Leibniz Institute for Astrophysics Potsdam, DE • May 2025: Celebration of John Bally: Star Formation, Stellar Feedback and the Ecology of Galaxies, Visegrád, HU • Oct 2023: Harvard–Heidelberg Star Formation Workshop, Center for Astrophysics, Cambridge MA, US • Nov 2021: 7th Australia–China Workshop on Astrophysics (ACAMAR7), AU	
Awards, Honors and Grants	SDSS-V Early Career Researcher Travel Grant 2026 CSIRO Space and Astronomy Research Studentship 2021 Swinburne University CfAS Research Studentship 2020 Ida C. Koran Student Aid Scholarship Program (Undergraduate) 2018 Cambridge GCE A-Level High Achiever Award 2017 Jeffery Cheah Entrance Scholarship 2016	
Organising Committee	• <i>LOC</i>, GMAC2: The Global Malaysian Astronomers Convention 2027 • <i>SOC</i>, European Astronomical Society Annual Meeting, ECR Session 2024, 2025 • <i>Co-chair</i>, Institute for Theoretical Astrophysics Journal Club 2023 • <i>LOC</i>, International Astronomical Union Symposium (IAUS) 377 2022	
Work/Teaching	<i>Teaching Assistant</i> , University of Heidelberg, Germany • Introduction to Computational Physics 2023 • MVComp: Fundamentals of Simulation Methods 2022 <i>Duty Astronomer</i> , Australia Telescope Compact Array, Australia Jan 2022 <i>Tutor</i> , Self-Employed, Australia 2018 - 2021	
Observing Experience	Australia Telescope Compact Array Jan 2022 • Studying radio candidates from the FLASH survey. <i>[6 hours]</i>	
Outreach and Mentoring	• <i>Writer</i>, Astrobites — the Astro-ph Reader’s Digest Sep 2021 <i>The mystery of orphan H II regions: where are the ionising star clusters?</i> • <i>President</i>, Science Society at Kuen Cheng High School, Malaysia 2015	
References	• Ralf Klessen klessen@uni-heidelberg.de Full Professor, University of Heidelberg • Kathryn Kreckel kathryn.kreckel@uni-heidelberg.de University of Heidelberg • Mark Krumholz mark.krumholz@anu.edu.au Full Professor, Australian National University	